

Lowered methionine ingestion as responsible for the decrease in rodent mitochondrial oxidative stress in protein and dietary restriction possible i...

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Available information indicates that long-lived mammals have low rates of reactive oxygen species (ROS) generation and oxidative damage at their mitochondria. On the other hand, many studies have consistently shown that dietary restriction (DR) in rodents also decreases mitochondrial ROS (mtROS) production and oxidative damage to mitochondrial DNA and proteins. It has been observed that protein restriction also decreases mtROS generation and oxidative stress in rat liver, whereas neither carbohydrate nor lipid restriction change these parameters. This is interesting because protein restriction also increases maximum longevity in rodents (although to a lower extent than DR) and is a much more practicable intervention for humans than DR, whereas neither carbohydrate nor lipid restriction seem to change rodent longevity. Moreover, it has been found that isocaloric methionine restriction also decreases mtROS generation and oxidative stress in rodent tissues, and this manipulation also increases maximum longevity in rats and mice. In addition, excessive dietary methionine also increases mtROS generation in rat liver. These studies suggest that the reduced intake of dietary methionine can be responsible for the decrease in mitochondrial ROS generation and the ensuing oxidative damage that occurs during DR, as well as for part of the increase in maximum longevity induced by this dietary manipulation. In addition, the mean intake of proteins (and thus methionine) of Western human populations is much higher than needed. Therefore, decreasing such levels to the recommended ones has a great potential to lower tissue oxidative stress and to increase healthy life span in humans while avoiding the possible undesirable effects of DR diets.